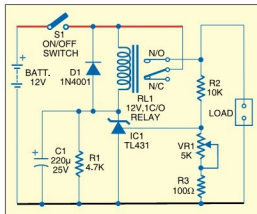


combination of preset VR1 and R3 that provide reference voltage. The battery voltage does not go below the reference voltage, preventing it from going into deep discharge.

The heart of the circuit is IC TL431 (IC1), which is a well-known voltage reference IC. The battery voltage is compared with the internal reference voltage of regulator IC1. As long as the voltage set by the preset VR1 is above the internal reference voltage, IC1 allows relay RL1 to energise, connecting the load to the battery. As the battery begins to discharge its terminal voltage decreases. When the battery voltage reaches the set voltage, say 10.5V, the cathode of IC1 goes high, which de-energises relay RL1 and disconnects the load from the battery.

Assemble the circuit on a general-purpose PCB and set the low cut-off



voltage (deep discharge voltage) of the battery. For setting the low cut-off voltage, remove the battery from the circuit and connect the circuit to a variable 12V DC power supply. Also connect a digital multimeter across the variable power supply. Increase the voltage from 0 to 12V through the variable power supply and read the

voltage in multimeter. Adjust preset VR1 in such a way that relay RL1 gets energised and load is connected to the battery. Now decrease the supply to 10.5V (low battery voltage setting recommended by manufacturers to avoid deep discharge). Move the preset VR1 slowly until the relay de-energises. You may try this adjustment more than once to get IC1 to switch at exactly 10.5V. After the adjustment is

done remove the variable power supply and multimeter, and reconnect the battery and switch on S1.

After calibration, enclose the assembled unit in a suitable shock-proof cabinet. Fix on/off switch S1 on the front panel and a two-pin connector for load on the back panel. Keep the battery inside the cabinet. ●

CAR REVERSE HORN

■ PRADEEP G.

This car reverse horn can be used with any vehicle having a 12V battery. Transistors T2 and T3 form an astable multivibrator (AMV) that generates beeping tone signals. Medium-power pnp transistor T1 functions as a simple speaker amplifier.

BD139 (T4) is an npn medium-power transistor. A 5.6V zener diode (ZD1) is connected to its base. ZD1 provides 5.6V DC regulated power supply to the horn circuit built around transistors T1 through T3. Diode D1 functions as reverse polarity protector. 12V DC to the circuit is connected via brake switch of the car/



three-wheeler.

When you reverse your vehicle, the circuit is connected to the battery of the car through the brake switch. The AMV oscillates and produces an audio frequency, which is given to speaker driver transistor T1 to generate

a beeping tone. This tone can be altered by changing the values of capacitors C2 through C4 in multivibrator circuit.

Assemble the circuit on a general-purpose PCB and enclose in a suitable box. Connect 12V DC power supply to the circuit through the brake switch. When you reverse your car the circuit gets 12V required for its working. Place the cabinet in the vehicle such that it produces sound at the rear while reversing the car. ●

